

Kathryn Caruso

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EDUCATION

2024 – Present	PhD MICROBIOLOGY AND CELL BIOLOGY Montana State University GPA: 4.00/4.00 Advisor: Christine FOREMAN, PhD
2025	MOLECULAR AND CELL BIOLOGY OF SYMBIOSIS Marine Biological Laboratory
2020	BSc BIOLOGY, Minor: RELIGIOUS STUDIES University of North Carolina, Asheville <i>summa cum laude</i> , GPA: 3.97/4.00 Advisor: Jonathan HORTON, PhD Thesis: “Assessing the effect of eastern hemlock (<i>Tsuga canadensis</i>) decline from hemlock woolly adelgid (<i>Adelges tsugae</i>) infestation on ectomycorrhizal colonization and growth of red oak (<i>Quercus rubra</i>) seedlings”

PEER-REVIEWED PUBLICATIONS

- Dieser, M., Smith, H. J., **Caruso, K.**, and C. M. Foreman. (2025). Seven genome sequences of bacterial environmental isolates from the western coast of the Greenland Ice Sheet. *Microbiology Resource Announcements*, e01192-25.
- Estep, M. C., Clark, L., Emery, K., Sanigepalli, S., **Caruso, K. E.**, Morgan, A., Greene, D., and J. R. Ward. (2025). Reproductive and genetic patterns in Virginia spiraea (*Spiraea virginiana* Britton; Rosaceae) reveal a fragmented species, frozen in place and time. *Castanea*, 90(1):48–66.
- Caruso, K. E.**, Horton, J. L., and A. A. Hove. (2021). Assessing the effect of eastern hemlock (*Tsuga canadensis*) decline from hemlock woolly adelgid (*Adelges tsugae*) infestation on ectomycorrhizal colonization and growth of red oak (*Quercus rubra*) seedlings. *American Midland Naturalist*, 186:16–34.

RESEARCH POSITIONS

2022 – 2024	North Carolina Center Lead and Communications Fellow <i>NOAA National Centers for Environmental Information, NASA DEVELOP</i> Designed and managed six research projects applying NASA Earth observations to environmental management decisions in collaboration with 14 academic, government, and non-profit organizations. Supervised six project teams of 3–5 researchers each. Supported Software Carpentry Workshops on Python and Git. Managed operations of NASA DEVELOP’s NOAA NCEI Office and supported career-building for early-career scientists.
2021 – 2022	Ecological Forecasting Researcher <i>NASA Ames Research Center & University of Georgia Center for Geospatial Research, NASA DEVELOP</i> Collaborated with federal, academic, and non-profit agencies to model present and future Venus flytrap habitat and monitor spongy moth defoliation using remote sensing imagery and environmental data.
2019 – 2020	Research Fellow <i>McCullough Institute</i>

Examined *Spiraea virginiana* reproduction and physiology to assess barriers to species' survival comparing reproductive effort and output among populations and testing pollen and seed viability through selfing, inbreeding, and outbreeding treatments. Facilitated plant education workshops for pre-K–middle school students. Collected and propagated *S. virginiana* for reintroduction and reintroduced only known member of extirpated *S. virginiana* population maintained in Arnold Arboretum Living Collections to Botanical Gardens at Asheville.

2018 **Hemlock Research Assistant**
 – 2020 *Southern Appalachian Cooperative Ecosystems Studies Unit, U.S. Park Service*
 Assessed efficacy of biological and chemical treatments of hemlock woolly adelgid by monitoring hemlock tree canopy health, hemlock woolly adelgid infestation rates, and *Laricobius* spp. predator beetle densities.

FUNDING & SCHOLARSHIPS

2025 **Crawford Wildlife Habitat Scholarship**
TOTAL: \$10,000
 2024 **National Science Foundation Extreme Biofilms National Research Traineeship**
TOTAL: \$100,000
 2024 **Mildred Livingston Grant Memorial Presidential Fellowship**
TOTAL: \$10,000
 2020 **French Broad Garden Club Scholarship**
TOTAL: \$1,250
 2020 **UNC Asheville Undergraduate Research Presentation Travel Grant**
TOTAL: \$500
 2019 **McCullough Fellowship**
TOTAL: \$5180
 2019 **UNC Asheville Undergraduate Research Grant**
TOTAL: \$804
 2016 **Laurels Fellowship**
TOTAL: \$19,000

AWARDS

2022 **NASA DEVELOP Performance Award**
 Recognized for supervising the Gatlinburg and Beatty Wildfires Project.
 2022 **NASA Ames Research Center DEVELOPER of the Term**
 Recognized for research on the Vermont and New Hampshire Ecological Forecasting Project.
 2020 **Bernhardt Perry Award for Undergraduate Research**
 Awarded for academic performance and contributions to four research projects.
 2020 **Manly E. Wright Award Nominee**
 Nominated for university-wide valedictorian award by the UNC Asheville Dept. of Biology.

NASA TECHNICAL REPORTS

1. Zhu, S., **Caruso, K. E.**, Dean, M., and J. Orser. (2022). Monitoring trends in tree defoliation due to *Lymantria dispar* outbreaks to predict future hardwood tree mortality and health impacts. *NASA Technical Reports Server*.
2. Rock, M., **Caruso, K. E.**, Lampley, J., Siddhi, A., and S. Zhu. (2021). Utilizing NASA Earth observations to map suitable Venus flytrap habitat in an effort to inform conservation, seed banking, and reintroduction in the Carolina Coastal Plain and Sandhills Regions. *NASA Technical Reports Server*.

PROFESSIONAL POSITIONS

2022 **Program Coordinator, AmeriCorps Summer of Service**
Conserving Carolina
 2022 **Forestry Technician**
Hemlock Restoration Initiative

2020	AmeriCorps Habitat Restoration Member
– 2021	<i>O'ahu National Wildlife Refuge Complex, Kupu Hawai'i Conservation Leadership Development Program</i>
2019	Science Writer
– 2020	<i>National Environmental Modeling and Analysis Center</i>

PRESENTATIONS

Oral Presentations:

1. **Caruso*, K. E.**, Madden, M., Payne, S., Rock, M., and A. Siddhi. (2023). Spatial ecological modeling of suitable habitat and forecasted land cover change using remote sensing data to identify at-risk populations of *Dionaea muscipula* (Venus flytrap) and inform species conservation efforts. *Annual Meeting, American Geophysical Union*: 11–15 December 2023, San Francisco, CA. — *Presenter
2. Allsbrook, K., Burke, S., **Caruso*, K. E.**, Childs-Gleason, L., Clayton, A., and K. Ross. (2023). 25 Years of the NASA DEVELOP Experiential Learning Model. *Annual Meeting, American Geophysical Union*: 11–15 December 2023, San Francisco, CA. — *Presenter
3. **Caruso, K. E.**, Lampley*, J., Rock, M., Siddhi, A., Zhu, S., Randall, J. L., Waitt, D., Weakley, A., Payne, S., and M. Madden. (2022). GIS spatial analysis and habitat suitability modeling using remote sensing data to help identify at-risk populations of US Carolina Coastal Plain endemic Venus flytrap (*Dionaea muscipula*). *Annual Meeting, Botanical Society of America*: 24–27 July 2022, Anchorage, AK. — *Presenter
4. **Caruso*, K. E.**, and J. L. Horton. (2020). Assessing the effect of eastern hemlock decline from hemlock woolly adelgid infestation on mycorrhizal fungi. *UNC Asheville 2020 Spring Undergraduate Research Symposium*: 21 April 2020, Asheville, NC. — *Presenter
5. **Caruso*, K. E.** (2019). Lessons in Leaves: Learning from Plants through Scientific and Contemplative Practice. *2019 McCullough Fellowship Annual Luncheon*: 4 December 2019, Asheville, NC. — *Presenter

Poster Presentations:

1. **Caruso*, K. E.**, Rock, M., Siddhi, A., Payne, S., and M. Madden. (2024). Informing *Dionaea muscipula* (Venus flytrap) Conservation through Habitat Modeling and Forecasted Land Cover Change. *oSTEM 14th Annual Conference*: 17–20 October 2024, Portland, OR. — *Presenter
2. Bolinger, G., **Caruso, K. E.**, Monteiro, T., Sgan*, A., Villalobos-Heredia, C., West, T., and M. Woloszyn. (2023). Utilizing NASA Earth observation data to monitor trends in air quality during a drought case study to improve public health response to drought threats. *American Geophysical Union Annual Meeting*: 11–15 December 2023, San Francisco, CA. — *Presenter
3. **Caruso*, K. E.**, and J. R. Ward. (2020). Factors Contributing to Reproductive Success and Failure in Virginia Spiraea (*Spiraea virginiana*), a Federally Threatened Shrub. *UNC Asheville 2020 Spring Undergraduate Research Symposium*: 21 April 2020, Asheville, NC. — *Presenter
4. **Caruso*, K. E.**, Kranyik, J., and J. R. Ward. (2020). Factors contributing to reproductive success and failure in Virginia spiraea (*Spiraea virginiana*), a federally threatened shrub. *81st Annual Meeting, Association of Southeastern Biologists*: 25–28 March 2020, Jacksonville, FL. — *Presenter, conference canceled following abstract acceptance due to COVID-19 pandemic
5. **Caruso*, K. E.**, Horton, J. L., and A. A. Hove. (2020). Assessing the effect of eastern hemlock decline from hemlock woolly adelgid infestation on mycorrhizal fungi. *81st Annual Meeting, Association of Southeastern Biologists*: 25–28 March 2020, Jacksonville, FL. — *Presenter, conference canceled following abstract acceptance due to COVID-19 pandemic

Invited Talk:

1. **Caruso*, K. E.** (2023). Highlights from NASA Capacity Building: DEVELOP. *Health and Air Quality Applications Program Review 2023*: 29–30 March 2023, Asheville, NC. — *Presenter

RESEARCH PROJECTS

- Biomaterials for Stabilizing Defense Materials in Arctic Environments
Montana State University Sept. 2024 – *Present*
- Measuring Crop Yields in Northern Brazil During El Niño Conditions to Evaluate Trends in Agricultural Production and Support Crop Forecasting
NOAA National Centers for Environmental Information Jan. – March 2024
- Multi-Hazard Approach to Mapping Flood Susceptibility and Vulnerability in Kentucky
NOAA National Centers for Environmental Information Jan. – March 2024
- Methods for Monitoring Rhode Island Habitats: Contributing to a Framework for Targeted Conservation and Management
NOAA National Centers for Environmental Information Jan. – March 2024
- Monitoring Trends in Air Quality During a Drought Case Study to Improve Public Health Response to Drought Threats
NOAA National Centers for Environmental Information June – Aug. 2023
- Using Earth Observations to Evaluate Snow Variability through a Climatological Analysis to Support Ecological Monitoring in Northeast Alaska
NOAA National Centers for Environmental Information Jan. – March 2023
- Evaluating the Role of Soil Moisture in Determining Vegetation Health, Fuel Loads, and Wildfires in the Gatlinburg and Beatty Wildfires
NOAA National Centers for Environmental Information Sept. – Nov. 2022
- Monitoring Trends in Tree Defoliation Due to *Lymantria dispar* Outbreaks to Predict Future Hardwood Tree Mortality and Health Impacts
NASA Ames Research Center Jan. – March 2022
- Utilizing NASA Earth Observations to Map Suitable Venus Flytrap Habitat in an Effort to Inform Conservation, Seed Banking, and Reintroduction in the Carolina Coastal Plain and Sandhills Regions
UGA Center for Geospatial Research Sept. – Nov. 2021
- Assessing the effect of eastern hemlock (*Tsuga canadensis*) decline from hemlock woolly adelgid (*Adelges tsugae*) infestation on ectomycorrhizal colonization and growth of red oak (*Quercus rubra*) seedlings
Depts. of Biology, UNC Asheville and Warren Wilson College May 2019 – May 2020
- Factors contributing to reproductive success and failure in Virginia spiraea (*Spiraea virginiana*), a federally threatened shrub
Dept. of Biology, UNC Asheville May 2019 – Aug. 2020
- A CEREUS (Consortium Exchanging Research Experiences for Undergraduate Students) Approach to Assessing Ecological Responses in the Southern Appalachians to Environmental Change
Dept. of Biology, UNC Asheville Jan. 2019 – May 2020
- Hemlock Assessment, Southern Appalachian Cooperative Ecosystems Studies Unit
Carl Sandburg Home National Historic Site, U.S. Park Service May 2019 – May 2020

CERTIFICATIONS

- Software Carpentry Instructor 2023

SERVICE

- Boardroom Bobcats Board Member, Gallatin Watershed Council Aug. 2025 – *Present*
- Volunteer, MSU Science Day Feb. 2025 – Feb. 2026
- Treasurer, Montana State University oSTEM Chapter Dec. 2024 – Nov. 2025
- STEM education and outreach, Emily Dickinson Elementary School Jan. – April 2025
- Volunteer, MSU MicroFair March 2025
- Volunteer, MSU Microbiology & Cell Biology Career Workshop Feb. 2025

MEDIA

- **Ignite@AGU:** Queering in the Earth and ecological sciences. Dec. 2023
https://www.youtube.com/watch?v=kGia83Mg7ts&list=PL8X9E6I5_i8g0-Dj6Aixg3AwPVx_gfdIA&index=9&pp=iAQB
- **Original Minds** of UNC Asheville interview April 2021
<https://www.youtube.com/watch?v=BGYgc8Xn3r8>
- **McCullough Fellows:** Working locally for global impact Feb. 2020
<https://stories.unca.edu/mccullough-fellows>